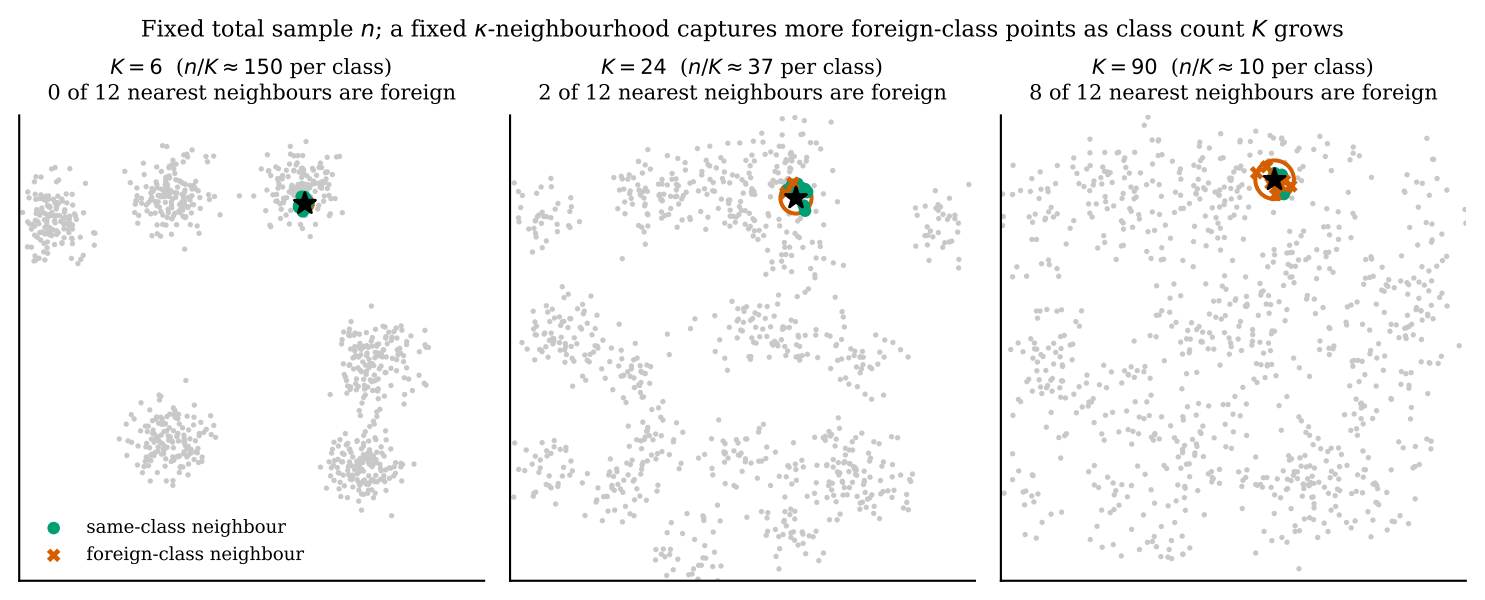

Fixed total sample n; a fixed κ-neighbourhood captures more foreign-class points as class count K grows
$K = 6$ ($n/K \approx 150$ per class)
0 of 12 nearest neighbours are foreign
$K = 24$ ($n/K \approx 37$ per class)
2 of 12 nearest neighbours are foreign
$K = 90$ ($n/K \approx 10$ per class)
8 of 12 nearest neighbours are foreign
same-class neighbour
foreign-class neighbour